ETL Process Documentation (For Part 2_Task 2B.pdf)

ETL Strategy for DigiHealth Modernization

1. Extraction: > Data is extracted from the normalized MySQL OLTP source tables (*Patients, Doctors, Appointments, MedicalRecords, and Billing*). This stage captures real-time transactional updates while maintaining data integrity via `ON DELETE CASCADE` actions.

2. Transformation: > Data is transformed from its 3NF structure into a dimensional format suitable for a **Star Schema**. Key transformations include:

- Concatenating `FirstName` and `LastName` into a single `FullName` field in `DimPatients`.
- Standardizing `Status` and `PaymentStatus` fields for categorical analysis.
- Aggregating billing amounts into a `Revenue` metric within the `FactAppointments` table.
- Assigning Surrogate Keys (SERIAL) to handle slowly changing dimensions.

3. Loading: > The transformed data is loaded into the PostgreSQL Data Warehouse. Dimension tables are populated first to satisfy referential integrity, followed by the Fact table which links all dimensions via foreign keys. This supports high-performance querying for healthcare metrics like "Revenue by Specialization" or "Patient Volume by Diagnosis."

Full SQL Script (Include in Part 2_Task 2B.pdf)

Below is the consolidated script required for your report.

OLTP Schema (MySQL)

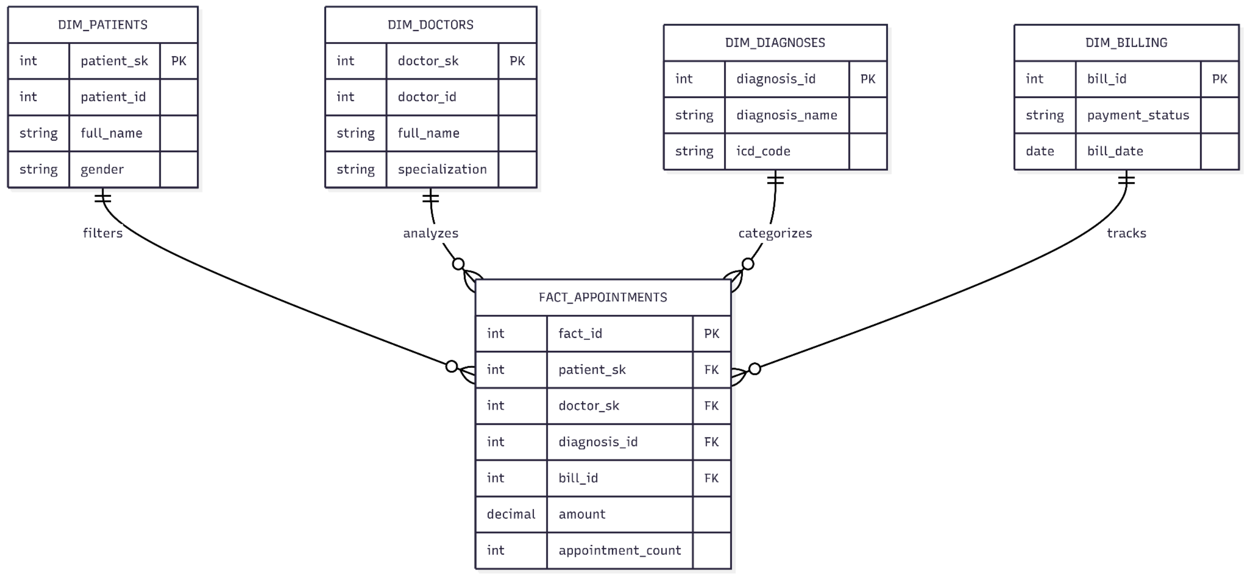
```
SQL
-- CREATE TABLE SCRIPTS
CREATE TABLE Patients (
   PatientID INT AUTO_INCREMENT PRIMARY KEY,
   FirstName VARCHAR(50) NOT NULL,
   LastName VARCHAR(50) NOT NULL,
   DOB DATE NOT NULL,
   Gender VARCHAR(10),
   ContactNumber VARCHAR(15)
);
```

```
CREATE TABLE Doctors (
    DoctorID INT AUTO_INCREMENT PRIMARY KEY,
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50) NOT NULL,
    Specialization VARCHAR(100)
);

CREATE TABLE Appointments (
    AppointmentID INT AUTO_INCREMENT PRIMARY KEY,
    PatientID INT,
    DoctorID INT,
    AppointmentDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    Status VARCHAR(20) CHECK (Status IN ('Scheduled', 'Completed', 'Cancelled')),
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID) ON DELETE CASCADE,
    FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID) ON DELETE CASCADE
);

CREATE TABLE MedicalRecords (
    RecordID INT AUTO_INCREMENT PRIMARY KEY,
    PatientID INT,
    DoctorID INT,
    Diagnosis TEXT,
    Prescription TEXT,
    VisitDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID) ON DELETE CASCADE,
    FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID) ON DELETE CASCADE
);

CREATE TABLE Billing (
    BillID INT AUTO_INCREMENT PRIMARY KEY,
    PatientID INT,
    Amount DECIMAL(10,2) NOT NULL,
    PaymentStatus VARCHAR(20) CHECK (PaymentStatus IN ('Paid', 'Pending', 'Overdue')),
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID) ON DELETE CASCADE
);

-- INSERT SCRIPTS
INSERT INTO Patients (FirstName, LastName, DOB, Gender) VALUES ('John', 'Doe',
'1985-05-10', 'Male');
INSERT INTO Doctors (FirstName, LastName, Specialization) VALUES ('Jane', 'Smith',
'Cardiology');
INSERT INTO Appointments (PatientID, DoctorID, Status) VALUES (1, 1, 'Completed');
INSERT INTO MedicalRecords (PatientID, DoctorID, Diagnosis) VALUES (1, 1, 'I10 - Essential
Hypertension');
```

INSERT INTO Billing (PatientID, Amount, PaymentStatus) VALUES (1, 200.00, 'Paid');

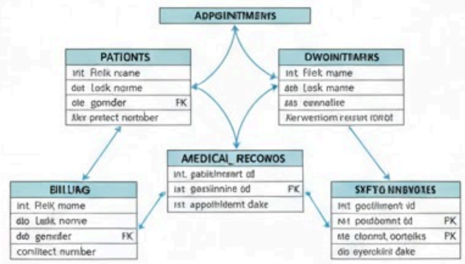
OLAP Schema (PostgreSQL)

```sql
SQL
-- DIMENSIONS
CREATE TABLE DimPatients (PatientID SERIAL PRIMARY KEY, FullName VARCHAR(100),
Gender VARCHAR(10));
CREATE TABLE DimDoctors (DoctorID SERIAL PRIMARY KEY, FullName VARCHAR(100),
Specialization VARCHAR(100));
CREATE TABLE DimDiagnoses (DiagnosisID SERIAL PRIMARY KEY, DiagnosisName TEXT);
CREATE TABLE DimBilling (BillID SERIAL PRIMARY KEY, PaymentStatus VARCHAR(20));

-- FACT TABLE
CREATE TABLE FactAppointments (
   AppointmentID SERIAL PRIMARY KEY,
   PatientID INT REFERENCES DimPatients(PatientID),
   DoctorID INT REFERENCES DimDoctors(DoctorID),
   DiagnosisID INT REFERENCES DimDiagnoses(DiagnosisID),
   BillID INT REFERENCES DimBilling(BillID),
   Revenue DECIMAL(10,2)
);

-- INSERT SCRIPTS
INSERT INTO DimPatients (FullName, Gender) VALUES ('John Doe', 'Male');
INSERT INTO DimDoctors (FullName, Specialization) VALUES ('Jane Smith', 'Cardiology');
INSERT INTO DimDiagnoses (DiagnosisName) VALUES ('I10 - Essential Hypertension');
INSERT INTO DimBilling (PaymentStatus) VALUES ('Paid');
INSERT INTO FactAppointments (PatientID, DoctorID, DiagnosisID, BillID, Revenue) VALUES
(1,
```



DIM_PATIENTS
int patient_sk PK
int patient_id
string full_name
string gender
DIM_DOCTORS
int doctor_sk PK
int doctor_id
string full_name
string specialization
DIM_DIAGNOSES
int diagnosis_id PK
string diagnosis_name
string icd_code
DIM_BILLING
int bill_id PK
string payment_status
date bill_date
filters
analyzes
categorizes
tracks
FACT_APPOINTMENTS
int fact_id PK
int patient_sk FK
int doctor_sk FK
int diagnosis_id FK
int bill_id FK
decimal amount
int appointment_count

Part 1, Task 1: DigiHeallth OLTP Schema (3NF)



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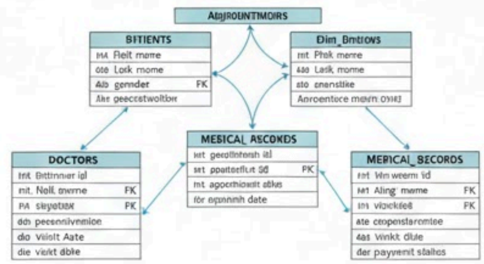

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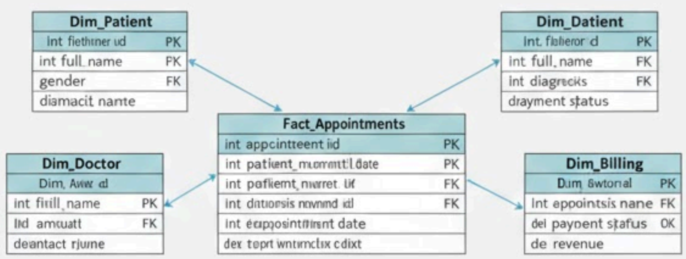
Part 2, Task 2: OLAP OLAP Schema



Explaation for Report

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